$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\frac{\partial f}{\partial x} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \frac{\partial^2 f}{\partial x^2} \quad \frac{\partial^3 f}{\partial x^2 \partial y} \quad \frac{\partial^3 f}{\partial x \partial y \partial x} \quad \frac{\partial}{\partial t} \quad \frac{\partial^2 u}{\partial x_1^2}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} \end{bmatrix}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} & \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_3^2} \end{bmatrix}$$

$$H_g = \begin{bmatrix} \dfrac{\partial^2 g(v)}{\partial y_1^2} & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_n} \\ \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_2^2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 g(v)}{\partial y_n \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_n \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_n^2} \end{bmatrix}$$

$$H_f(\mathbf{a}) = \begin{bmatrix} \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1^2} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2^2} \end{bmatrix} \qquad H_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Proposition 1 (Identity)**.** *For every $n \times n$ matrix A we have $I_n A = A$.*

Exercise 2. Compute $\begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$ for $f(x_1, x_2) = x_1^2 x_2$.

By **Proposition 1**, **Exercise 2** needs no identity matrix. See also **Theorem 4.2** in the course notes.